GLOSSARY

UNIT 01: CELL CYCLE : MITOSIS, AND MEIOSIS

1. Cell: The basic unit of life in all living organisms.

2. Nucleus: The control centre of a cell that contains genetic material (DNA).

3. Cytoplasm: The jelly-like substance inside a cell where organelles are suspended.

4. Chromosomes: Thread-like structures made of DNA that carry genetic information.

5. Genetic: Related to genes and heredity.

6. Mitosis: The process of cell division that produces two identical daughter cells.

7. Meiosis: The type of cell division that produces gametes (sperm and egg cells) with half the number of chromosomes.

8. Cytokinesis: The final stage of cell division where the cytoplasm splits, forming two separate cells.

9. Significance: The importance or meaning of something.

10. Phase: A distinct stage in a process, such as the phases of cell division.

11. Repairing: The process of fixing or restoring damaged cells or tissues.

12. Healing: The biological process of recovering from injury or illness.

13. Reproduction: The process by which organisms produce offspring.

14. Gametes: Reproductive cells (sperm and egg) that carry genetic information for offspring.

UNIT 02: HOMEOSTATIS:

1. Homeostasis: The process of maintaining a stable internal environment in the body.

2. Feedback: A biological mechanism that helps regulate processes in the body, such as temperature or hormone levels.

3. Internal: Located inside the body or within an organism.

4. Regulation: The control or maintenance of a process, often to keep balance in the body.

5. Transpiration: The process of water loss from plants through small openings in leaves.

6. Hydration: The process of absorbing or maintaining an adequate amount of water in the body.

7. Optimal: The best or most favourable condition for function or performance.

8. Enzyme: A protein that speeds up chemical reactions in the body.

9. Shivering: An involuntary body response that generates heat by muscle contractions to maintain body temperature.

10. Dehydration: A condition caused by excessive loss of water from the body.

UNIT 03: TRANSPORT IN LIVING ORGANISMS:

1. Transport: The movement of substances, such as nutrients and gases, within an organism.

2. Photosynthesis: The process by which plants convert sunlight into energy, producing oxygen and glucose.

3. Pathway: A route or process through which substances move in the body.

4. Blood: A fluid in the body that transports oxygen, nutrients, and waste materials.

5. RBCs (Red Blood Cells): Cells in the blood that carry oxygen using haemoglobin.

6. WBCs (White Blood Cells): Cells in the blood that help fight infections and protect the body.

7. Platelets: Small blood cells that help in clotting to prevent excessive bleeding.

8. Plasma: The liquid part of the blood that carries cells, nutrients, and waste products.

9. Veins: Blood vessels that carry blood back to the heart, usually deoxygenated.

10. Arteries: Blood vessels that carry oxygenated blood away from the heart to the body.

11. Deoxygenated: Blood that has low oxygen content, usually returning to the heart.

12. Oxygenated: Blood that is rich in oxygen, usually leaving the heart to supply the body.

UNIT 04: FUNDAMENTALS TO CHEMISTRY:

1. Chemistry: The branch of science that studies the composition, structure, properties, and reactions of matter.

2. Inorganic: The study of compounds that do not contain carbon-hydrogen (C-H) bonds, such as metals and minerals.

3. Organic: The study of carbon-containing compounds, primarily associated with living organisms.

4. Physical: The branch of chemistry that deals with the physical properties and behavior of matter, including energy changes.

5. Biochemistry: The study of chemical processes and substances that occur in living organisms.

UNIT 05: CHEMISTRY TERMINOLOGIES

1. Atom: The smallest unit of matter that retains the properties of an element.

2. Molecule: A group of two or more atoms bonded together.

3. Nucleus: The central part of an atom that contains protons and neutrons.

4. Proton: A positively charged subatomic particle found in the nucleus of an atom.

5. Neutron: A neutral (uncharged) subatomic particle found in the nucleus of an atom.

6. Electron: A negatively charged subatomic particle that orbits the nucleus of an atom.

7. Elements: Pure substances consisting of only one type of atom.

8. Compound: A substance formed when two or more different elements chemically bond together.

9. Heterogeneous: A mixture in which the components are not evenly distributed and can be distinguished.

10. Homogeneous: A mixture in which the components are evenly distributed and appear uniform throughout.

11. Chemical Equation: A symbolic representation of a chemical reaction using chemical formulas.

UNIT 05: CHEMICAL REACTIVITY:

1. Metals: Elements that are typically shiny, good conductors of heat and electricity, and malleable.

2. Non-Metals: Elements that are usually dull, poor conductors of heat and electricity, and brittle.

3. Metalloid: Elements that have properties of both metals and non-metals.

4. Conductor: A material that allows electricity or heat to pass through it easily.

5. Alkaline Metals: Highly reactive metals found in Group 1 of the periodic table, such as sodium and potassium.

6. Halogens: Highly reactive non-metals found in Group 17 of the periodic table, such as fluorine and chlorine.

7. Reactivity: The tendency of a substance to undergo a chemical reaction, either by itself or with other substances.

12. **Reactant:** The starting substances in a chemical reaction that undergo change.

13. **Product:** The substances formed as a result of a chemical reaction.

UNIT 07: INTRODUCTION TO KINEMATICS:

1. **Kinematics:** The branch of physics that studies the motion of objects without considering the forces causing it.

2. **Motion:** The change in position of an object over time.

3. **Force:** A push or pull acting on an object that can cause a change in motion.

4. **Translatory:** Motion in which all parts of an object move in the same direction and distance.

5. **Rotatory:** Motion in which an object moves in a circular path around a fixed axis.

6. **Vibratory:** Motion in which an object moves back and forth around a fixed point.

7. **Scalar Quantities:** Physical quantities that have only magnitude (size) but no direction, such as mass and speed.

8. **Vector Quantities:** Physical quantities that have both magnitude and direction, such as velocity and force.

9. **Distance:** The total length of the path travelled by an object, regardless of direction.

10. **Speed:** The rate at which an object covers distance, measured as distance divided by time.

11. **Average Velocity:** The total displacement divided by the total time taken.

12. **Displacement:** The shortest distance between an object's initial and final position, with direction.

UNIT 08: INTRODUCTION TO DYNAMICS:

1. Dynamics: The branch of physics that studies the forces and their effects on motion.

2. Force: A push or pull acting on an object that can cause it to accelerate or change direction.

3. Mass: The amount of matter in an object, usually measured in kilograms (kg).

4. Velocity: The speed of an object in a given direction.

5. Acceleration: The rate at which an object's velocity changes over time.

6. Deceleration: Negative acceleration causing a decrease in velocity over time.

7. Friction: A force that opposes the motion of an object when it is in contact with a surface.

8. Newton: The SI unit of force, named after Sir Isaac Newton.

$$(1 \text{ Newton} = 1 \text{ kg}\cdot\text{m/s}^2)$$

9. Weight: The force exerted by gravity on an object, calculated as: **mass × gravitational acceleration ($W = mg$).**

10. Inertia: The tendency of an object to resist changes in its state of motion unless acted upon by an external force.

UNIT 09: FORCES AND THEIR EFFECTS:

1. Hydraulic Lifts: Machines that use pressurized fluid to lift heavy objects by applying Pascal's principle.

2. Stretched: The action of elongating or extending an object due to an applied force.

3. Deformation: The change in shape or size of an object due to the application of a force.

4. Stiffness: The resistance of an object to deformation when a force is applied.

5. Pressure: The force applied per unit area $(P = F/A)$.

6. Area: The surface over which a force is applied or distributed.

7. Fluid: A substance that can flow, such as a liquid or gas, and has no fixed shape.

8. Density: The mass of an object per unit volume $(\rho = m/V)$.

9. Depth: The distance from the surface to a lower point, often affecting pressure in fluids.